Sub-100 lb Pumpkin Trebuchet

Elastic-powered trebuchet, total machine weight under 100 lb, sized to throw a one-pound pumpkin more than 510 feet — design basis, weight budget, bill of materials, build instructions, tuning, and safety.

1. Why this is not a counterweight trebuchet

A one-pound pumpkin is about 4 in. in diameter, and at the speeds involved **air drag dominates its flight**. Simulating a 0.454 kg, 10 cm sphere (C_d 0.47) released 10 ft above ground:

Release speed	Best range (≈40° launch)
150 ft/s (45 m/s)	≈ 410 ft
165 ft/s (50 m/s)	≈ 465 ft
180 ft/s (55 m/s)	≈ 515 ft
200 ft/s (60 m/s)	≈ 570 ft

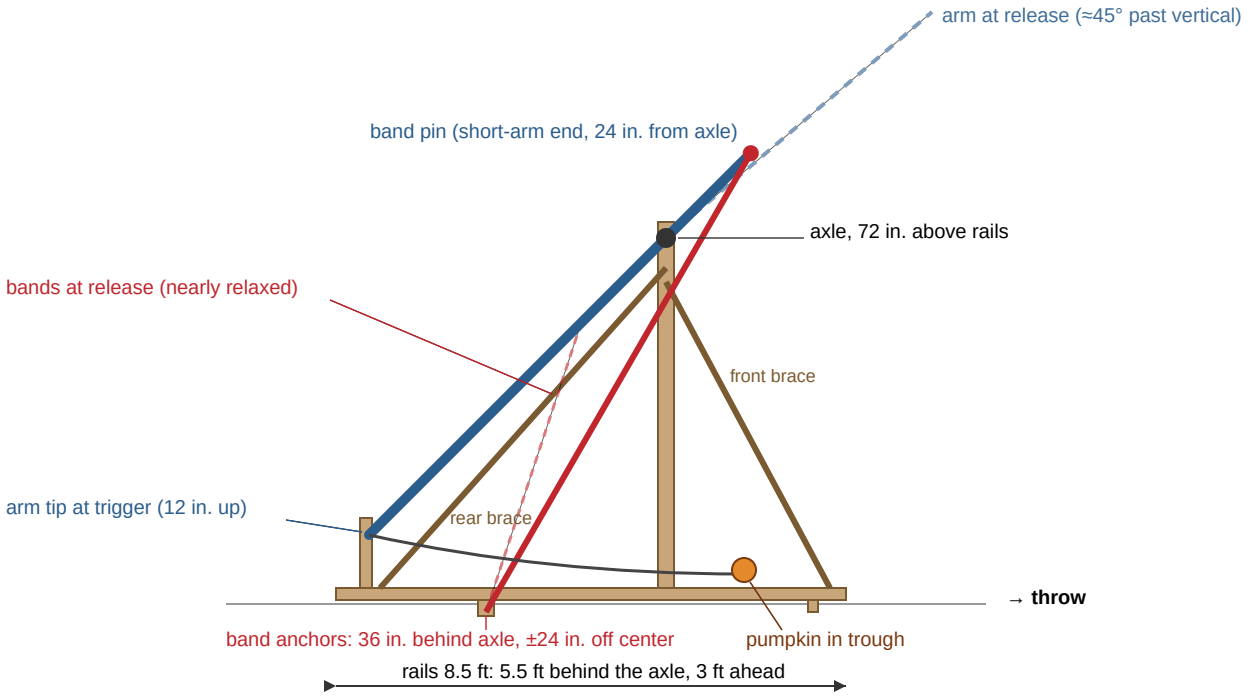
Clearing 510 ft with margin needs **roughly 190–200 ft/s at release**, which is about 800 J of projectile kinetic energy. A good trebuchet puts 40–55% of its stored energy into the projectile, so the machine must store **about 2,000–2,500 J**.

> **Gravity cannot supply that inside a 100 lb total weight limit.** Even if the frame and arm were only 40 lb, the remaining 60 lb of counterweight dropping 5 ft stores about 400 J. At 55% efficiency that gives ~110 ft/s and a **range of about 250 ft**. Reaching 510 ft by gravity needs a 350–400 lb counterweight.
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> The energy must therefore come from **elastic storage**: natural-latex tubing stores roughly 2–4 kJ per kg, so 2–3 lb of latex holds the 2,000–3,000 J required. The machine below is a conventional arm-and-sling trebuchet in every respect except that the short arm is pulled by latex bands anchored to the rear of the base instead of by a falling weight.
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> **If your competition rules forbid springs or elastics**, the only other energy source that is not counted in machine weight is human pullers (a traction trebuchet). With 6–8 pullers, 300–400 ft is realistic for a 1 lb pumpkin; 510 ft is not a reliable expectation.

Machine specification

Type	Elastic (latex) powered, fixed axle, sling release	**Target weight**	≈ 93 lb (budget below); limit 100 lb
Throwing arm	9 ft aluminum, two-piece, 10.4 lb	**Pivot location**	2 ft from the band end (7 ft long arm, 3.5 : 1)

Axle height	72 in. above the base rails	**Cocked arm angle**	45° below horizontal, tip to the rear
Energy storage	8–12 latex bands, ½ in. ID × ¾ in. OD × 36 in.	**Band stretch**	290% cocked (≈104 in.), ~70% at release
Peak band force	400–960 lbf total, depending on band count	**Sling**	≈ 6–6.5 ft, adjustable
Frame footprint	8.5 ft long × 4.5 ft wide (outriggers)	**Projectile**	1 lb pumpkin, ≈3.5–4 in. diameter



Side view, to scale (1 ft = 30 px). Solid arm = cocked; dashed = at release. Red = latex bands (one bundle each side of the frame). Posts, braces, and bands sit outside the arm's swing plane.

2. Weight budget

Estimated with kiln-dried SPF lumber (2×4 ≈ 1.1 lb/ft, 2×3 ≈ 0.8, 2×2 ≈ 0.55, 1×2 ≈ 0.4) and 6061-T6 aluminum. Lumber varies ±15% with species and moisture, so **weigh every stick before cutting** and use the trim options if the running total creeps up.

Component	Spec	lb
Arm, main section	2 × 3 × ⅛ in. 6061-T6 rectangular tube, 60 in.	7.0
Arm, tip section	1.5 × 1.5 × ⅛ in. 6061-T6 square tube, 54 in.	3.4
Axle + bearings	1 in. steel × 20 in. + 2 flange bearings (UCFL205-16)	7.2
Base rails	2 × (2×3 × 8.5 ft)	13.6
Posts	2 × (2×3 × 75 in.)	10.0
Rear braces	2 × (2×2 × 86 in.) + 2 × (1×2 mid-ties, 30 in.)	9.9
Front braces	2 × (2×2 × 69 in.)	6.3
Rear anchor beam / outrigger	2×6 × 54 in.	7.6
Front outrigger	1×4 × 48 in.	2.2
Lateral braces	2 × (1×2 × 60 in.)	4.0
Trough	¼ in. plywood 13.5 × 96 in. + 2 × 1×2 guide strips	6.2
Trigger post, tabs, pin	2×3 × 14 in., two steel tabs, ½ in. quick-release pin	2.5
Latex bands	12 × 36 in., ½ in. ID × ¾ in. OD	3.6
Band hardware	4 steel rings, 2 quick links, 2 eye bolts, ¾ in. band pin	3.5
Fasteners, stakes	Bolts, screws, 4 ground stakes	5.0
Sling, pouch, release pin, cords		1.5
Total (estimated)		**93.5**

Trim options if you are over: rear anchor beam 2×6 → 2×4 with a steel strap doubler (−2.5 lb); trough plywood ⅛ in. hardboard (−2 lb); run 8 bands instead of 12 (−1.2 lb); rails 2×3 → 2×2 with 1×4 face plates at the post joint (−3 lb); shorten the trough to 6 ft (−1.5 lb). Do not trim the arm wall thickness, the axle, or the band hardware.

3. Bill of materials

3.1 Aluminum and steel

Qty	Item	Use
1	6061-T6 rectangular tube, 2 × 3 × ⅛ in. wall, 60 in.	Arm main section (band end through pivot to 36 in. past axle)

1	6061-T6 square tube, 1.5 × 1.5 × ⅛ in. wall, 54 in.	Arm tip section (6 in. inserted into main tube)
2	⅛ × 2 × 4 in. aluminum plate	Doubler plates at the axle hole
1	1 in. dia. 1018 cold-rolled steel round, 20 in.	Axle
2	2-bolt flange bearing, 1 in. bore (UCFL205-16)	Axle bearings, bolted to post outer faces
2	Spacer tube, 1 in. ID × ~1¼ in. OD, 5¾ in. long (steel or aluminum)	Center the arm between the posts
1	⅜ × 3½ in. grade 8 bolt, nylock nut	Cross-pin locking arm to axle
1	¾ × 8 in. grade 8 bolt, nylock nut, 2 washers	Band pin through the short-arm end
4	Welded steel ring, 2 in. ID × ½ in. stock (≥2,000 lb)	Band collector rings (2 on the arm pin, 2 at the anchors)
2	½ in. quick link, rated ≥3,000 lb	Anchor ring to eye bolt
2	½ × 6 in. forged shoulder eye bolt, nut, fender washer, 3 × 3 × ⅛ in. plate washer	Band anchors in rear beam
1	½ × 4 in. eye bolt, nut	Trigger eye near arm tip
1	¼ × 3 in. eye bolt, nut	Fixed sling cord attachment
1	5/16 in. steel rod, 6 in.	Sling release pin
1	½ in. quick-release (ball-lock or hitch) pin with lanyard ring	Trigger pin
2	⅛ × 1½ × 6 in. steel flat bar, drilled ½ in.	Trigger tabs on trigger post
2	⅜ × 4 in. bolts, nylock nuts	Arm joint (tip tube to main tube)
4	½ × 3½ in. bolts, nuts, washers	Flange bearings through posts
4	12 in. steel stakes or ½ in. rebar pins	Anchor outriggers to the ground

3.2 Lumber (kiln-dried SPF, straight, no large knots)

Qty	Item	Cut to / use
2	2×3 × 10 ft	Base rails, 102 in.; offcut → trigger post 14 in.
2	2×3 × 8 ft	Posts, 75 in.
2	2×2 × 8 ft	Rear braces, 86 in.
2	2×2 × 6 ft	Front braces, 69 in.
1	2×6 × 5 ft	Rear anchor beam / outrigger, 54 in.
1	1×4 × 4 ft	Front outrigger, 48 in.
2	1×2 × 8 ft	Lateral braces, 60 in. each; offcuts → rear brace mid-ties 30 in.

2	1×2 × 8 ft	Trough guide strips
1	¼ in. plywood, 13.5 × 96 in. strip	Trough
1	Hardwood (oak/maple) block, 1¾ × 2¾ × 8 in.	Plug for the band-pin end of the arm (4 in.) and shims at the arm joint
1	Hardwood block, 1¼ × 1¼ × 4 in.	Plug in arm tip for the release pin

3.3 Energy storage, sling, cordage

Qty	Item	Use
40 ft	Natural latex (amber) tubing, ½ in. ID × ¾ in. OD (⅛ in. wall)	12 bands × 36 in. + spares. Buy from one lot.
1 spool	Waxed polyester whipping twine, ~1 mm	Binding band ends to rings
25 ft	⅛ in. (3 mm) Dyneema/Amsteel cord, ≥1,400 lb	Sling cords (light cords matter with a 1 lb projectile)
1	6 × 5 in. ballistic nylon or leather, 4 small grommets	Pouch
1	1 in. ID welded ring, 3/16 in. stock	Sling release ring
60 ft	3/16 in. cord	Trigger pull cord
30 ft	⅜ in. rope + 2 small pulleys (3:1 haul)	Cocking line
1	Hanging crane / luggage scale, 110 lb (220 lb better)	Measuring band force (see 5.1)

3.4 Fasteners

Qty	Item	Use
16	⅜ × 3½ in. hex bolts, nuts, washers	Posts to rails, braces to posts and rails, anchor beam to rails
1 lb	2½ in. structural screws	Outriggers, lateral braces, trough blocks
½ lb	1¼ in. screws	Trough and guide strips
1	Structural epoxy (e.g., JB Weld or West System)	Hardwood plugs in arm tubes

3.5 Projectiles

One-pound pumpkins: **‘Wee-B-Little’** or small **‘Baby Pam’** reach about 1 lb at 3.5–4 in. diameter. Choose the roundest, densest, smoothest fruit; for the same weight, a smaller diameter flies farther. Approximate total cost: $350–500.

4. Build instructions

Coordinate convention used below: **x** is forward (direction of throw), measured from the axle; heights are above the top of the base rails. The two posts and two rails are mirror pairs about the arm's swing plane.

4.1 Base

1. Cut the two 2×3 rails to 102 in. Lay them on edge, parallel, with **12 in. clear between them** (13.5 in. center to center). They run from x = −66 in. (rear) to x = +36 in. (front).
2. Bolt the 2×6 rear anchor beam flat across the underside of both rails at x = −36 in., centered, so it projects 27 in. each side. Two ⅜ in. bolts per rail.
3. Screw the 1×4 front outrigger across the underside at x = +30 in., projecting 24 in. each side.
4. Install the two ½ in. shoulder eye bolts vertically through the anchor beam at ±24 in. from center, eyes on top, with the 3 in. plate washers underneath. The eyes should sit just outboard of the rail line and about level with the rail tops.
5. Cut a 14 in. trigger post from 2×3 offcut and bolt it between the rails at x = −60 in. (its top at 14 in.). Bolt the two steel trigger tabs to its top so they stand up 2 in. with their ½ in. holes aligned across the arm plane, spaced 1¼ in. apart (the trigger eye bolt will sit between them).

4.2 Posts and braces

1. Cut two 2×3 posts to 75 in. Bolt one to the outer face of each rail at x = 0, 3.5 in. dimension running fore-aft, with two ⅜ in. bolts through post and rail. Posts are now 13.5 in. apart (clear).
2. Rear braces (2×2 × 86 in.): from the outer face of each post at 66 in. height down to the rail top at x = −57 in. Bolt each end with one ⅜ in. bolt through brace, post (or rail) and a 1×2 scab on the far side. Add a 1×2 mid-tie from the brace midpoint to the post at 36 in. height (screws) to halve the brace's buckling length.
3. Front braces (2×2 × 69 in.): from the post outer face at 63 in. height down to the rail top at x = +33 in., bolted the same way.
4. Lateral braces (1×2 × 60 in.): from each post's outer face at 48 in. height down and outward to the ends of the front outrigger. Screw both ends. The frame should now be rigid in all directions; rack it by hand and add a screw anywhere it moves.
5. Drill the ½ in. bearing bolt holes through each post so the axle centerline is at **72 in.** above the rail tops. Mount the flange bearings on the post outer faces, bolts through the post. Slide the axle in dry to check that the two bearings align; shim one bearing if the axle binds.

4.3 Throwing arm

1. Epoxy the 4 in. hardwood plug into one end of the 2×3 tube — this is the **band end**. Measure 1.5 in. from that end and drill a ¾ in. hole through the two 3 in. faces (the hole runs side to side when the arm is in the frame). This takes the band pin.
2. Measure **24 in.** from the band end and drill the 1 in. axle hole through the same pair of faces, centered on the 3 in. dimension. Epoxy and clamp the two ⅛ in. aluminum doubler plates over the hole on each face (drill them to match) — then redrill 1 in. through everything.

3. Make the arm joint: cut hardwood shims so the 1.5 in. square tube sits snugly inside the far end of the 2×3 tube, centered. Insert the tip tube 6 in., and drill two ⅜ in. holes through both tubes and shims (at 1.5 in. and 4.5 in. from the main tube's end) for the joint bolts. Total arm length should now be 108 in.: band end → axle 24 in., axle → tip 84 in.
4. Epoxy the 1¼ in. hardwood plug into the tip and drill it axially for the 5/16 in. release pin, which should project 3 in. and be a snug push fit (add a cross set-screw so it cannot walk out). Drill for the ¼ in. sling eye bolt 2 in. from the tip and the ½ in. trigger eye bolt 5 in. from the tip, both through the lateral faces.
5. Install the arm: slide the axle through one bearing, one spacer tube, the arm, the second spacer, and the second bearing. Cross-drill the arm and axle together with ⅜ in. and install the grade 8 cross-pin. Tighten the bearing set screws onto the axle. The arm must swing freely and the tip must be about 12 in. above the rails when the arm points 45° down toward the rear.

4.4 Latex bands

1. Cut twelve 36 in. bands from one lot of tubing. Bands within a set should be the same length within ⅛ in.
2. Make two upper bundles and two lower bundles. For each band end: pass 3 in. of tubing through a 2 in. ring, fold it back on itself, and bind the doubled section with 30+ tight wraps of waxed twine finished with a constrictor knot, then a second binding ½ in. further along. Never clamp latex with hose clamps or zip ties — they cut it.
3. Start with 4 bands per side (8 total): each band runs from an upper ring to a lower ring. Hang the two upper rings on the ¾ in. band pin, one each side of the arm, with a washer between ring and arm and a nylock nut outside. Connect each lower ring to its eye bolt with a ½ in. quick link.
4. With the arm standing vertical the bands should be slack-to-lightly tensioned (about 60 in. between pin and eye). When the arm is cocked (tip at the trigger) they stretch to about 104 in., which is ~290% elongation — the maximum you should ever use.

4.5 Sling, trough, trigger, cocking

1. Pouch: hem the 6 × 5 in. panel, set a grommet at each corner, and thread one Dyneema cord through the two grommets on each short side. Tie one cord permanently to the ¼ in. eye bolt; splice the 1 in. release ring onto the other. Start with an effective sling length (tip to pouch center) of **72 in.** and leave the ring end adjustable.
2. Bend the release pin so it points about **40° forward** of the arm axis (toward the throw). The ring drops off this pin at release.
3. Trough: screw the ¼ in. plywood strip onto the rail tops from x = −60 in. to +36 in., passing between the posts. Screw the two 1×2 guide strips on top to form a **5.5 in. wide channel** on the centerline. Wax the channel.
4. Cocking: tie the ⅜ in. rope to the trigger eye bolt, run it through a pulley lashed to the trigger post base and back through a second pulley on the arm (3:1). Two people haul the tip down and drop the ½ in. quick-release pin through tabs and eye. **Install a safety lashing** (a loop of rope from the arm tip around the trigger post) before anyone approaches to load the pouch.
5. Loading: lay the sling forward along the trough, set the pumpkin in the pouch at about x = +15 in., check the ring is seated on the release pin, remove the safety lashing, walk back 50 ft, and pull the trigger cord.

5. Tuning

5.1 Measure your bands first

Latex compounds vary by supplier, so measure instead of trusting a catalog number. Hang one 36 in. band from a beam with the crane scale in series, pull it to 104 in. (290% stretch), and read the force `F` in lbf. Approximate stored energy per band is:

`E (joules) ≈ 0.55 × F × 68 in. ÷ 12 × 1.356 ≈ 4.2 × F`

Measured F at 290%	Per band	8 bands	10 bands	12 bands
50 lbf	≈ 210 J	1,700 J	2,100 J	2,500 J
65 lbf	≈ 275 J	2,200 J	2,750 J	3,300 J
80 lbf	≈ 335 J	2,700 J	3,400 J	4,000 J

Target 2,200–3,000 J stored. The frame, arm, and hardware above are sized for a maximum of **12 bands at 80 lbf (960 lbf total)**. Do not exceed that.

5.2 Firing sequence

1. Dry-fire with 4 bands and an empty pouch to check that the sling runs clean and the arm stops safely. After release the bands re-stretch and brake the arm; it will oscillate for a few seconds before settling upright. Stay clear until it stops.
2. Throw a 1 lb sandbag in a sock at 6, then 8 bands. Adjust in this order, one variable at a time, three throws each:
   - **Release pin angle** until the projectile leaves at about 40° (flat and short → bend the pin forward; too steep or lands behind → bend it back).
   - **Sling length**: lengthen in 3 in. steps until range stops improving or the pouch drags; expect 72–80 in.
   - **Band count**: add bands in pairs (one each side) until you have 560+ ft on sand. Expect this between 8 and 12 bands.
3. Switch to pumpkins only after the machine is consistent on sandbags. Pumpkins vary; pick the densest ones.
4. If range plateaus, the next gain is reducing tip inertia: lighter sling cords, a smaller pouch, or a thinner-wall tip tube — not more bands.

6. Safety — non-negotiable

- **Range:** 700 ft clear downrange, 150 ft clear to each side, and **nobody in front of the axle while the trigger pin is in.** A release failure sends the pumpkin anywhere in a 180° arc, including straight up or backward.
- **Stored elastic energy is the main hazard.** A cocked machine holds up to ~3,000 J in bands pulling with up to 960 lbf. A band or binding that fails snaps back toward the arm pin at over 100 ft/s. Nobody stands beside the frame, in line with the bands, or reaches between the posts while the

machine is cocked. Use the safety lashing whenever hands are near the arm. Never leave the machine cocked unattended.

- Fire from 50 ft behind the machine using the pull cord. Everyone wears eye protection.
- Inspect every band and binding before each session; retire any band with nicks, chalky surface, or permanent set. Latex degrades in sunlight and ozone: store bands in a dark, cool bag and expect to replace them every season.
- Inspect the arm joint bolts, the axle cross-pin, the band pin, the anchor eye bolts, and the trigger tabs before each session. Check that the stakes are driven and the frame does not rock.
- Check local laws and event rules. Many jurisdictions treat this as a projectile device and require landowner permission and a cleared range; many events restrict or ban stored-elastic energy, so confirm before building.

Design basis: analytical estimates and a point-mass drag simulation. The band energy, arm stress, and frame load figures are engineering estimates with safety factors of 2–3; the measurement step in 5.1 and the incremental band-count procedure in 5.2 are how those estimates get verified on your actual materials before full power.